

Kumbara by Sembol

• TESTNET

A self-custodial dollar piggy bank for Türkiye, opened with a passkey: Face ID, Touch ID or a password manager.

Live (Stellar TESTNET)

kumbara.sembol.xyz

Repository

github.com/keyboord01/kumbara

Deck date

6 September 2026

Ahmed · Instaward grantee · author of [@sembol/passkey-react](#) · contributor to [stellar/smart-account-kit](#)

PROBLEM

Lira savings lose value; the dollar route has a wall in front of it.

Inflation makes lira saving a losing move

Consumer prices in Türkiye rose **31.51% year on year** in August 2026.¹ Dollar-denominated saving is how ordinary households defend themselves.

The crypto route exists but sheds users at every step

USDC on Stellar settles in seconds for fractions of a cent. Getting there means an extension or app install, twelve words to keep safe, XLM for gas bought somewhere else, and a phone you must never lose.

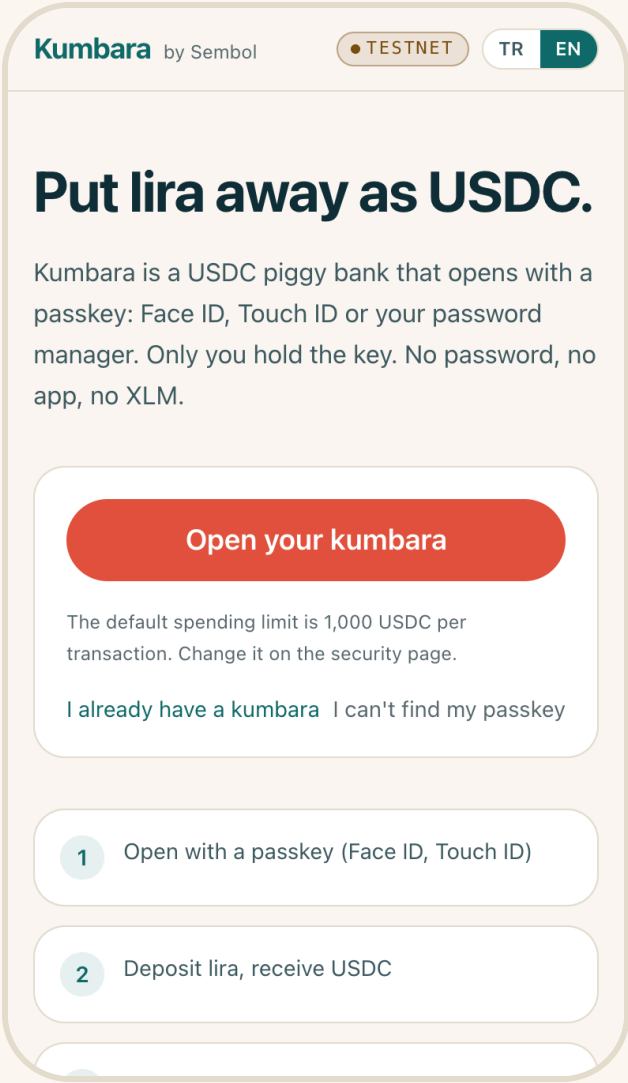
The people who need hard-currency savings most are served least

A bank-transfer habit and a phone with a passkey (Face ID or Touch ID) are what people have. That is the whole interface Kumbara asks for: no seed phrase, no XLM, no install.

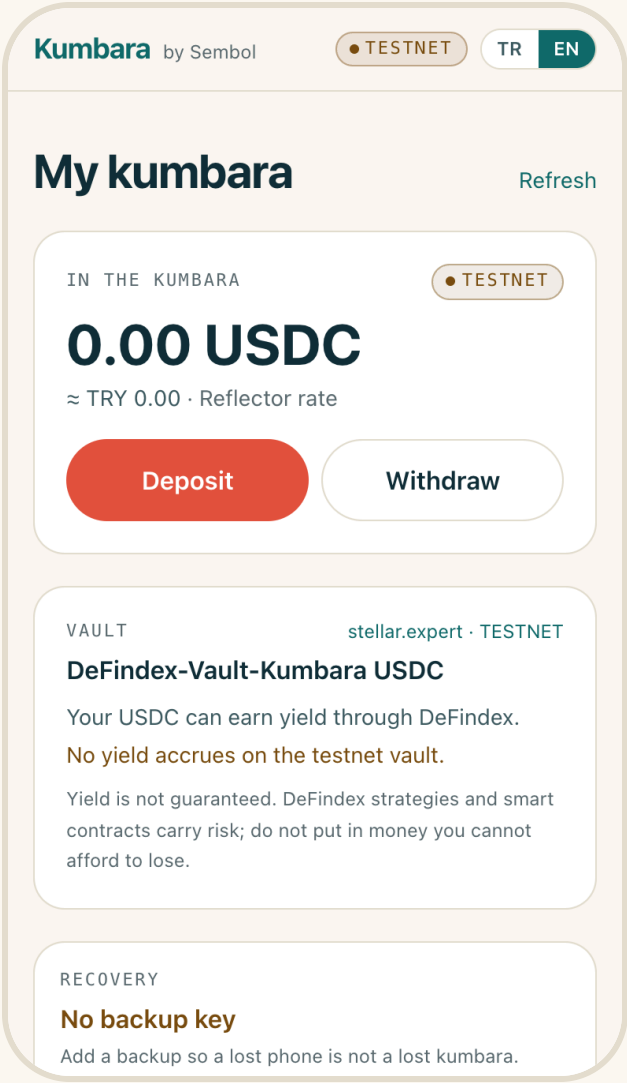
Türkiye is market one. The product is built for a class of markets: high-inflation economies with a bank-transfer habit and an existing Stellar anchor. The anchor is configuration, not code, so the same app serves the next such market.

¹ TÜİK (Turkish Statistical Institute), *Tüketici Fiyat Endeksi, Ağustos 2026*, news bulletin of 3 September 2026: CPI +31.51% against August 2025 (monthly +1.84%). Listed on tuik.gov.tr ("%31,51 (yıl)") with the bulletin at veriportali.tuik.gov.tr/tr/press/58290; verified on tuik.gov.tr on 6 September 2026.

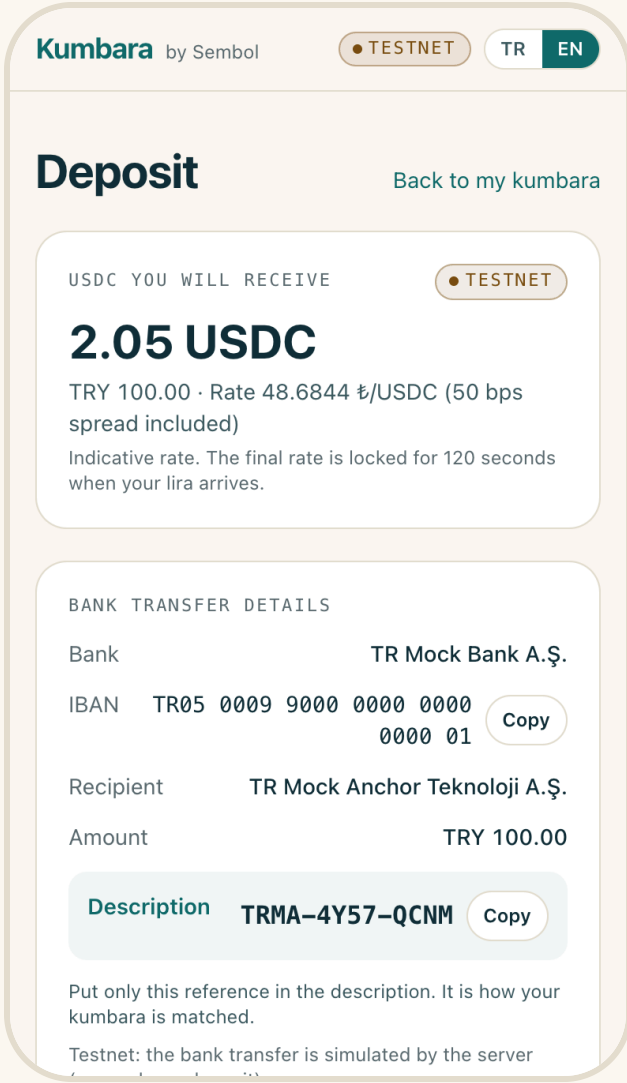
Four screens. Tap to account: ~10 s. No seed phrase, no XLM, no install.



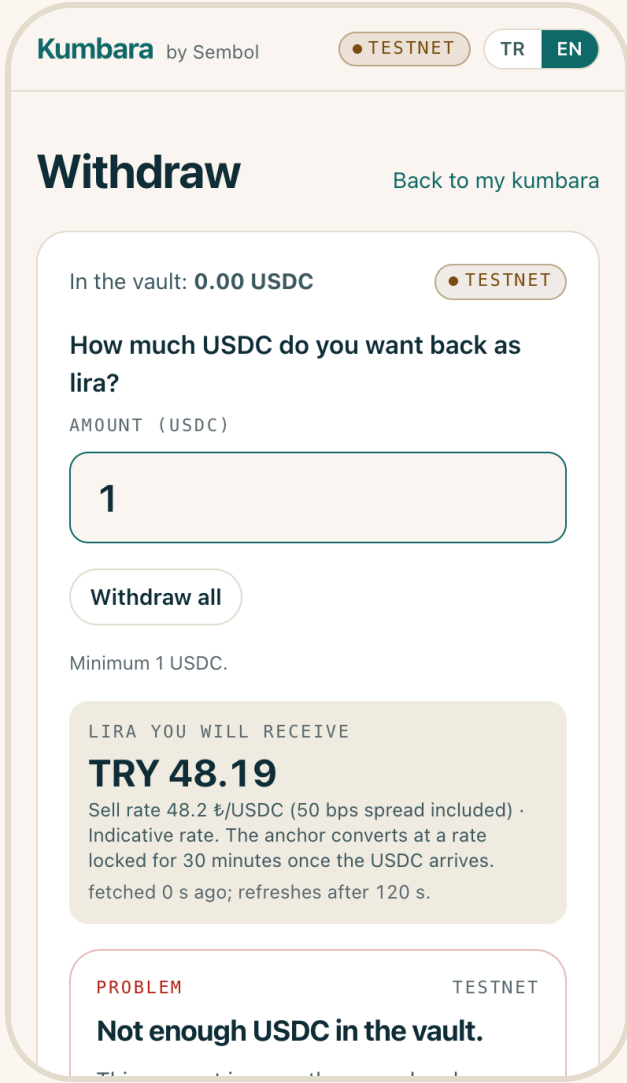
Onboard. One button; a passkey (Face ID, Touch ID, password manager) creates the key and the relay deploys an OpenZeppelin smart account. Measured 8–16 s tap to Savings in browser runs.



Savings. USDC in the DeFindex vault with its lira equivalent (Reflector rate), spending limit, recovery. Every figure carries the TESTNET badge.



Deposit. IBAN, recipient and a reference code, exactly like an online bank transfer; the anchor converts lira to USDC.



Withdraw. A sell quote in lira; two passkey approvals; the anchor pays the visitor's own IBAN.

Scan, tap once, and you have a kumbara before this slide is over.



kumbara.sembol.xyz ● TESTNET

1. Scan the code (it opens the app with this booth's reference).
2. Tap **Kumbaranı aç** and approve with Face ID or Touch ID: an OpenZeppelin smart account is deployed for you; the relay pays the fees.
3. Tap **Yükle**, enter 100, and the presenter plays the bank: about a minute later the USDC is in the DeFindex vault.
4. Tap **Çek** to send it back as lira; the payout reference appears in under a minute.

iPhone: Safari, not private browsing. Android: Chrome with a screen lock. Nothing to install, nothing to write down, no XLM. The counter on the booth screen and kumbara.sembol.xyz/stats move as you go.

HOW IT WORKS

Seven steps, one flow: anchor in front, DeFindex behind, a fee-sponsoring relay underneath.

1

The visitor opens an OpenZeppelin smart account with a passkey (Face ID, Touch ID); the relay pays the fees.

2

A spending limit (1,000 USDC per transaction) is written on-chain in the background.

3

The anchor issues an IBAN and a reference; the visitor sends lira (simulated by the server on testnet).

4

The anchor converts the lira to USDC and pays an ownerless, pre-authorized bridge account.

5

The bridge account forwards the USDC to the kumbara and closes itself; nobody holds its key.

6

One passkey approval puts the USDC into the DeFindex vault (no yield on testnet).

7

A withdrawal runs the path backwards: vault → bridge account → anchor → lira to the IBAN.

The full sequence diagram, every message from the passkey to the payout, is in docs/architecture.md in the repository.

Anchors pay classic accounts. Smart accounts are contracts. Ramp payouts could not reach them.

The finding

The anchor's on-ramp rejects a contract (C...) destination and its watcher matches classic payments only: a payout cannot land in a smart account, and a smart account's Soroban transfer is never matched on the way back.

The solution: ownerless, pre-authorized bridge accounts

A throwaway classic account whose only valid transactions are signed before it exists: the **forward** at sequence +1 (USDC to the user's smart account, the exact quoted amount) and the **cleanup** at sequence +2 (trustline dropped, reserves back to the sponsor). The master key is wiped in-process: **no key can sign anything else**. In reverse, the same primitive carries withdrawals to the anchor's treasury with the memo id.

10 / 10

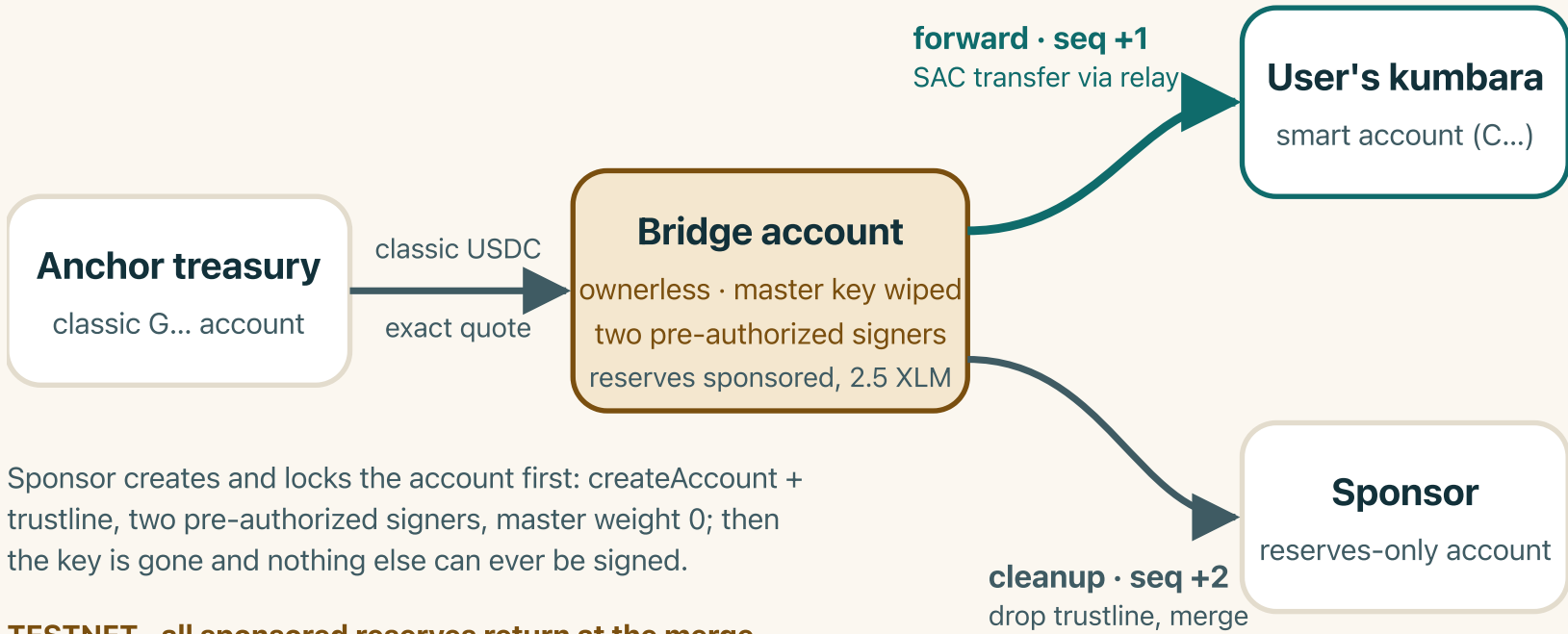
exact amount matches in ten scripted deposits

900

stroops net per deposit for the sponsor; reserves return at cleanup

4

signatures the bridge key ever produces, asserted by a unit test



Sponsor creates and locks the account first: createAccount + trustline, two pre-authorized signers, master weight 0; then the key is gone and nothing else can ever be signed.

TESTNET · all sponsored reserves return at the merge

EVIDENCE (STELLAR TESTNET, STELLAR.EXPERT/EXPLORER/TESTNET/TX/...)	TRANSACTION
Anchor pays the bridge account (classic payment)	778215b3...e3c3a8f1
Pre-authorized forward: USDC lands in the smart account	f580e030...f22c9eab
Pre-authorized cleanup: trustline removed, account merged	6223c61c...7416291d

Reported to the anchor team with the two fixes that would retire the bridge: SAC delivery to contract wallets and Soroban-aware matching on the way back. Full hashes in docs/anchor-notes.md.

Listed building blocks, wired end to end. Kumbara writes no wallet, vault or strategy code.

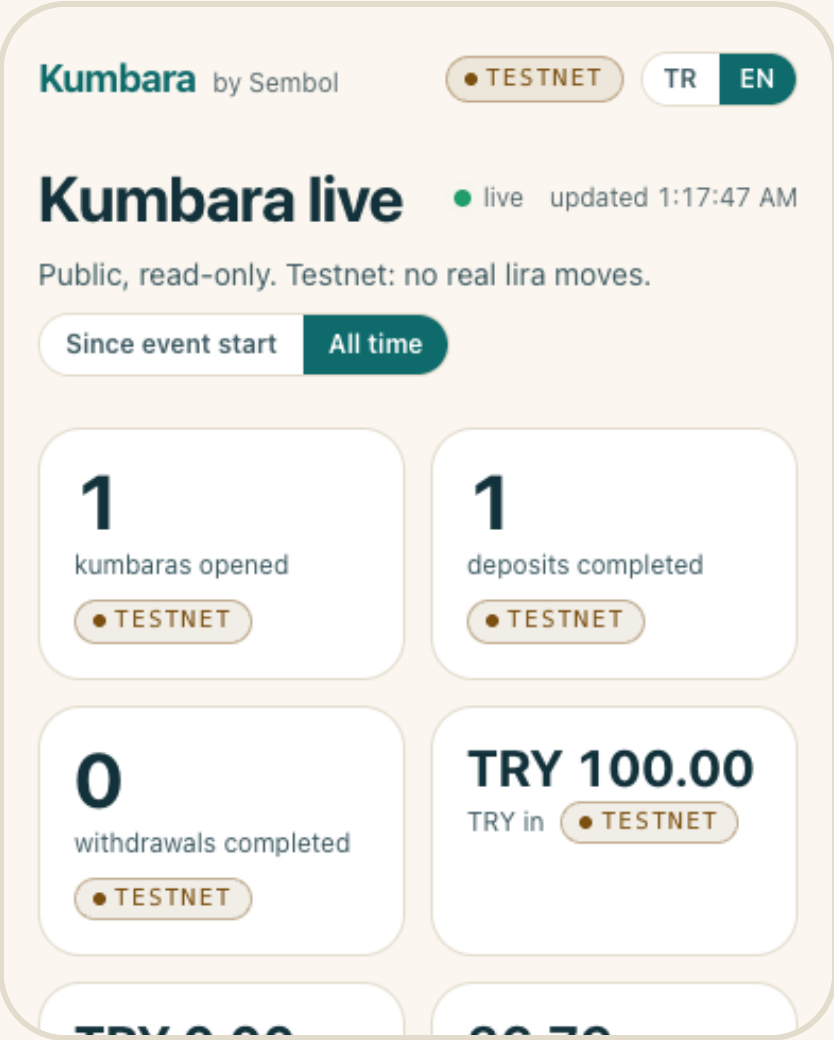
BUILDING BLOCK	WHAT IT DOES IN THIS APP	ADDRESS (TESTNET, STELLAR.EXPERT)
OpenZeppelin smart accounts + spending-limit policy	Every kumbara is an audited smart account whose only signer is the visitor's passkey; a USDC-scoped rule caps 1,000 USDC per transaction on-chain. Via @sembol/passkey-react and stellar/smart-account-kit.	WASM 1b5f4534...97785a · policy CABXBYJN...2TIP5G
DeFindex vault + hodl strategy	Deposits go through the vault's real invest path into its strategy and withdrawals unwind from it. The strategy is DeFindex's own hodl crate, built unmodified; no yield on testnet, none promised.	vault CAT76PQM...MKSKL · strategy CDFLOE4H...JLDMQN
TR Mock Anchor · SEP-1/6/38 + Partner API	The regulated lira side: IBAN and reference, quotes with a 120 s lock, on-ramps to the bridge account, off-ramps by memo, simulated FAST payouts. Issuer and endpoints read from its stellar.toml.	USDC SAC CBIELTK6...XQDAMA
Fee-sponsoring relay (OpenZeppelin Channels today)	Signed authorization entries go to our server route, which adds the project key and forwards to the relay; the visitor never sees XLM. Per-IP and per-booth caps, sponsor threshold.	channels.openzeppelin.com/testnet
Reflector*	The USD/TRY foreign-exchange oracle behind the lira equivalent on the Savings and stats pages (read by simulation, anchor rate as fallback).	CBKGPWGK...4DXMCJZC (mainnet oracle, read only)*

Upstream: "Fix wallet creation crash on default Next.js apps" was merged into [stellar/smart-account-kit](#) on 19 August 2026 and shipped in 0.6.2 · [github.com/stellar/smart-account-kit/pull/4](#)

Open-source plan: the repository is MIT-licensed. Kumbara deploys no proprietary contracts: the smart account, WebAuthn verifier and spending-limit policy are OpenZeppelin's audited contracts, the vault and strategy are DeFindex's; the bridge-account logic ([lib/landing](#)) is MIT in the repo with its tests.

* Reflector is a mainnet price oracle read by simulation only: Kumbara sends no mainnet transaction and makes no mainnet claim; every balance here is Stellar TESTNET and the anchor's quote is the fallback rate.

Everything on this slide is public, linkable and re-runnable.



kumbara.sembol.xyz/stats: headline numbers with a TESTNET badge, live feed, onboarding curve, timings. The *since event start* / *all time* toggle counts from 19 September 08:00 and shows all-time until then; seed and E2E accounts are excluded by default.

6

real-browser round trips in CI: onboard, deposit, withdraw, booth, stats, failure screens · every push and every 6 h against production

25

failure states, each a TR/EN screen with one action; every flow resumes from stored state

94 · 100 · 100 · 100

Lighthouse mobile on /stats: performance, accessibility, best practices, SEO

● E2E · passing · latest production run 6 Sep 2026

GitHub Actions, workflow e2e.yml (repository badge available once the repo is public)

ON-CHAIN EVIDENCE (STELLAR TESTNET)

TRANSACTION

Vault deployed through the DeFindex factory with the hodl strategy attached

dbf0d5b7...6faa8f96

Deposit with invest=true moved USDC into the strategy (invest)

b615e7dc...76b7f981

Withdrawal unwound the shares from the strategy (divest)

466f48d2...07a9f5ca

Recorded fallback demo: docs/demo/round-trip.mp4, the full round trip on kumbara.sembol.xyz, 2:39 with a caption per step.

WHAT 'S NEXT

- Sembol Cloud v0 on mainnet (Instaward Month 3)
- Anchor partnership conversations underway
- SCF Integration Track after the event